

**Leveraging Artificial Intelligence Technology in
Medical Case Study: Risks of Diabetic
Retinopathy**

Marvin Azuogu

Houston City College

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Professor: Dr. Doreen Rosenstrauch

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Case Study Summary

This report examines a clinical case involving artificial intelligence (AI) assisted screening for diabetic retinopathy, demonstrating how modern technology can identify health risks before they manifest as symptoms. The patient, Mrs. Maria Santos (pseudonym), is a 58-year-old woman who presented for routine diabetes management at her primary care clinic. Despite having no vision related complaints, AI screening revealed significant eye disease that required immediate ophthalmologic attention.

Patient Information and Presentation

Mrs. Santos has a 12-year history of Type 2 Diabetes Mellitus with suboptimal glycemic control, as evidenced by her hemoglobin A1c (HbA1c) level of 8.2%, which is above the recommended target of less than 7% for most adults with diabetes. She also has controlled hypertension and a body mass index (BMI) of 31, placing her in the obese category. Despite clinical guidelines recommending annual eye examinations for individuals with diabetes, Mrs. Santos had not undergone retinal screening in three years due to challenges accessing specialist appointments and transportation barriers.

During her primary care visit, fundus photographs were captured using a specialized retinal camera and analyzed by an FDA approved AI system called EyeArt. The AI algorithm processed the images within minutes and detected bilateral diabetic retinopathy, moderate nonproliferative diabetic retinopathy in the right eye and mild nonproliferative diabetic retinopathy in the left eye. This finding was particularly significant because Mrs. Santos reported no visual symptoms, illustrating how diabetic retinopathy can progress silently in its early and moderate stages (Rajesh et al., 2023).

Identified Health Risks

Based on the clinical presentation and AI-assisted diagnosis, three major health risks were identified for this patient:

Risk 1: Progression to Vision-Threatening Retinopathy and Blindness

Diabetic retinopathy is a leading cause of blindness among working-age adults globally, with approximately 103.12 million adults worldwide affected by this condition (Tahir et al., 2025).

The moderate stage detected in Mrs. Santos's right eye indicates that microvascular damage has already occurred in the retina, with hemorrhages, microaneurysms, and possibly cotton-wool spots present. Without intervention, moderate nonproliferative diabetic retinopathy has a significant likelihood of progressing to proliferative diabetic retinopathy, where abnormal new blood vessels grow on the retinal surface. These fragile vessels can hemorrhage into the vitreous cavity, causing sudden vision loss, and can lead to tractional retinal detachment, a sight threatening emergency.

This risk is particularly important because early-stage diabetic retinopathy is often asymptomatic, as demonstrated in this case. Patients typically do not notice vision changes until the disease has reached advanced stages when treatment options are more limited and outcomes are less favorable. Studies have shown that timely laser photocoagulation, anti VEGF injections, or other interventions can significantly reduce the risk of severe vision loss if implemented before the disease advances (Dai et al., 2024). If Mrs. Santos's condition goes untreated, she faces substantial risk of irreversible blindness within the next several years, which would dramatically impact on her quality of life, independence, and ability to manage her diabetes effectively.

Risk 2: Development of Diabetic Macular Edema

Diabetic macular edema (DME) represents another critical risk for Mrs. Santos. DME occurs when fluid accumulates in the macula, the central part of the retina responsible for sharp, detailed central vision. This complication can develop at any stage of diabetic retinopathy and is the most common cause of vision loss in patients with diabetes (Song & Song, 2024). Given Mrs. Santos's poor glycemic control and the presence of moderate retinopathy in one eye, she is at elevated risk for developing macular edema.

The importance of identifying this risk cannot be overstated. Unlike proliferative changes that primarily affect peripheral vision initially, macular edema directly impacts central vision, making it difficult for patients to read, drive, recognize faces, and perform activities of daily living. The presence of macular edema significantly worsens visual prognosis and requires more aggressive treatment approaches, including intravitreal injections of anti-vascular endothelial growth factor (anti-VEGF) medications. Early detection through AI screening allows for close monitoring and prompt intervention if macular thickening develops, potentially preserving Mrs. Santos's functional vision and independence.

Risk 3: Systemic Microvascular and Macrovascular Complications

The presence of diabetic retinopathy serves as a biomarker for systemic vascular damage beyond the eyes. The same microvascular pathophysiology that damages retinal blood vessels also affects other organ systems, placing Mrs. Santos at risk for diabetic nephropathy (kidney disease), diabetic neuropathy (nerve damage), and cardiovascular disease. Research has demonstrated strong correlations between the severity of retinopathy and the presence of other diabetes-related complications (Duggal et al., 2025).

This risk is important because it highlights that diabetic retinopathy is not merely an isolated eye problem but rather an indicator of widespread vascular injury. If Mrs. Santos's diabetic

retinopathy is left unaddressed alongside her poor glucose control, she faces increased risk of kidney failure requiring dialysis, peripheral neuropathy leading to foot ulcers and potential amputation, cardiovascular events such as heart attack and stroke, and premature mortality. The finding of moderate retinopathy should prompt comprehensive evaluation of other organ systems and intensification of her diabetes management to prevent or slow the progression of these life-threatening complications. The interconnected nature of diabetic complications means that protecting Mrs. Santos's vision through early intervention may also help preserve her kidney function, neurological health, and cardiovascular wellbeing.

Leveraging AI Technology to Manage and Reduce Risks

Artificial intelligence technology plays a transformative role in addressing each of these identified health risks through multiple mechanisms:

Early Detection and Intervention

The most significant contribution of AI-based screening is the ability to detect diabetic retinopathy at asymptomatic stages, as demonstrated in Mrs. Santos's case. Traditional screening models rely on patients scheduling and attending appointments with ophthalmologists or optometrists, which often result in significant delays or missed screenings altogether. AI systems like EyeArt, IDx-DR, and others can be deployed in primary care settings, enabling point-of-care screening during routine diabetes visits. These systems achieve sensitivity and specificity exceeding 85% for detecting referable diabetic retinopathy, comparable to human expert graders (Rajesh et al., 2023).

By detecting Mrs. Santos's moderate retinopathy three years after her last eye examination, the AI system prevented further disease progression that would have occurred during the gap in care. Early detection enables timely referral to ophthalmology for close monitoring, laser treatment if

needed, or anti-VEGF therapy for macular edema, all of which are more effective when implemented before advanced disease develops. Research demonstrates that AI screening programs significantly increase screening rates in underserved populations and reduce rates of vision-threatening complications through early intervention (Duggal et al., 2025).

Improved Access and Health Equity

Mrs. Santos's three-year gap in eye care exemplifies a widespread problem: many patients with diabetes face barriers to specialist eye care, including long wait times, geographical distance to ophthalmology practices, transportation challenges, and financial constraints. AI-based screening directly addresses these access barriers by bringing diagnostic capability to primary care clinics where patients already receive their diabetes management. Studies in rural and underserved areas have demonstrated that AI screening programs dramatically increase screening participation rates compared to traditional referral models (Uy et al., 2023).

This improved access is particularly important for reducing health disparities. Populations with limited access to specialty care, including rural residents, racial and ethnic minorities, and individuals with lower socioeconomic status, experience disproportionately high rates of diabetic blindness. By enabling screening where patients already receive care, AI technology helps ensure that vulnerable populations like Mrs. Santos receive timely diagnosis and treatment, potentially reducing vision loss disparities across different communities.

Risk Stratification and Personalized Monitoring

Emerging AI technologies go beyond simple detection to predict disease progression risk for individual patients. Deep learning algorithms can analyze retinal images to predict which patients are most likely to progress from mild to sight-threatening retinopathy within specific timeframes, allowing for personalized surveillance intervals (Dai et al., 2024). For Mrs. Santos,

AI analysis of her baseline images combined with clinical factors like her elevated HbA1c could predict her individualized risk of progression, enabling her ophthalmologist to schedule more frequent monitoring if she is at high risk or providing reassurance if her risk is lower.

This risk stratification capability helps optimize healthcare resource utilization by directing more intensive monitoring and preventive interventions toward patients who need them most, while potentially extending screening intervals for stable, low-risk patients. The ability to predict outcomes also facilitates shared decision-making between clinicians and patients about the intensity of glucose control efforts and the urgency of ophthalmologic follow-up.

Systematic Screening and Reduced Loss to Follow-Up

AI screening programs integrated into primary care workflows create systematic, protocol-driven screening that reduces the likelihood of patients falling through the cracks. Rather than depending on patients to remember to schedule separate eye appointments, AI screening can be automatically triggered during diabetes management visits. Electronic health record integration can track when screening is due and prompt clinicians to perform imaging, ensuring comprehensive population coverage.

For patients like Mrs. Santos who face logistical barriers to specialist care, immediate AI-based screening at the primary care visit ensures that detection occurs even if patients struggle to attend separate ophthalmology appointments. The AI system also generates automatic referrals for patients with referable disease, creating a systematic pathway from detection to specialist evaluation that reduces loss to follow-up compared to traditional screening models where patients receive referral instructions but may not complete the appointment (Song & Song, 2024).

Motivating Better Diabetes Management

Visual evidence of diabetic complications can serve as a powerful motivator for patients to improve their glucose control. When Mrs. Santos receives her AI screening results showing actual damage to her retinas from diabetes, this concrete demonstration of disease impact may motivate behavioral changes more effectively than abstract laboratory values. Research suggests that viewing one's own retinal images and understanding the findings can improve diabetes self-management behaviors and adherence to treatment regimens.

Additionally, knowing that regular screening will detect any worsening of her retinopathy may encourage Mrs. Santos to optimize her glucose control to prevent progression. The closed-loop feedback of 'improve control, see better screening results' creates positive reinforcement for diabetes management efforts. This behavioral aspect of AI screening complements the direct medical benefits, potentially reducing all microvascular and macrovascular risks through improved systemic diabetes control.

Conclusion: The Importance of Risk Identification in Healthcare

The case of Mrs. Santos powerfully demonstrates why systematic risk identification is a cornerstone of effective healthcare. Without AI screening, her moderate diabetic retinopathy would have remained undetected, likely progressing to vision-threatening disease over the subsequent months to years. The silent, asymptomatic nature of early and moderate diabetic retinopathy means that patients often do not seek care until irreversible damage has occurred. By identifying risks before symptoms develop, healthcare systems can intervene at stages when treatment is most effective and outcomes are most favorable.

Risk identification serves multiple critical functions in healthcare. First, it enables prevention through early intervention, as illustrated by Mrs. Santos's timely referral for ophthalmologic care before vision loss occurred. Second, it allows for resource allocation optimization, directing

intensive monitoring and treatment toward patients who need them most. Third, it facilitates patient education and engagement, empowering individuals to understand their health status and participate actively in disease management decisions. Fourth, it reduces health disparities by ensuring that vulnerable populations receive equitable access to diagnostic services.

The integration of artificial intelligence into risk identification processes represents a paradigm shift in how healthcare can be delivered. AI screening for diabetic retinopathy exemplifies how technology can democratize access to specialist-level diagnostics, bringing high-quality screening to underserved populations and primary care settings. As AI technologies continue to advance, their application will likely expand to other diseases where early, asymptomatic detection improves outcomes, including other diabetic complications, cardiovascular disease, certain cancers, and neurodegenerative conditions.

However, the successful implementation of AI-based risk identification requires thoughtful attention to the ethical considerations raised by Kong and Wang (2025) in their analysis of medical artificial intelligence. Healthcare systems must ensure that AI tools augment rather than replace human clinical judgment, that algorithmic outputs align with practical clinical value, and that the pursuit of efficiency through digitization does not erode the humanistic care that patients deserve. Clinicians must retain agency in interpreting AI results within the full clinical context of each patient's unique circumstances.

Ultimately, Mrs. Santos's case illustrates that the value of healthcare extends beyond treating established disease to include identifying and preventing future harm. Proactive risk identification through tools like AI-based diabetic retinopathy screening embodies the principle that preventing blindness is far preferable to treating it after it occurs. As healthcare systems worldwide grapple with rising rates of diabetes and other chronic diseases, the systematic

integration of risk identification into routine care, enhanced by artificial intelligence, will be essential to improving population health outcomes, reducing healthcare costs through prevention, and ensuring that all patients, regardless of their circumstances, have the opportunity to preserve their health and quality of life.

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